

1. Administrative & Study Identification

Full Study Title: A Phase 3b, Multicenter, Open-Label, Single-Arm Study of Acalabrutinib (ACP-196) in Subjects with Chronic Lymphocytic Leukemia **Short Title/Acronym:** ASSURE (Acalabrutinib Safety Study in Untreated and Relapsed or Refractory Chronic Lymphocytic Leukemia Patients) **Preferred UMLS Name:** Acalabrutinib Safety Study **Protocol Number:** D8220C00008 **NCT Number:** NCT04008706 **Posting ID:** 517912925405500954 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Acalabrutinib (Calquence) **Phase of Development:** Phase 3 (Phase 3b) **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** AstraZeneca Clinical Study Information Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of acalabrutinib, determined by the incidence and severity of adverse events. **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), Event-Free Survival (EFS), time to next treatment, pharmacokinetics, and patient-reported outcomes (e.g., EORTC-QLQ-C30).

2.2 Background & Rationale While Bruton’s tyrosine kinase (BTK) inhibitors have transformed the treatment landscape for Chronic Lymphocytic Leukemia (CLL), ongoing real-world and expanded clinical data is necessary to understand long-term safety, tolerability, and efficacy. The ASSURE study was initiated to prospectively investigate the safety and efficacy of acalabrutinib monotherapy across a broad patient population, including treatment-naïve patients, those with relapsed/refractory (R/R) disease, and patients who have previously been treated with other BTK inhibitors (specifically prior ibrutinib therapy).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 552 participants (Targeted Enrollment) **Number of Sites:** 135 global cancer research sites **Estimated Study Start:** 17-Sep-2019 **Estimated Primary Completion:** 04-Nov-2025 **Total Estimated Study Duration:** Approximately 72 months from the first participant enrolled.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Men and women ≥ 18 years of age. 2. Confirmed diagnosis of CLL meeting published diagnostic criteria (IWCLL 2018 criteria) with active disease requiring treatment. 3. ECOG performance status of 0, 1, or 2.

4.2 Exclusion Criteria 1. Primary malignancy other than CLL (except for adequately treated basal/squamous cell skin cancer or in situ cancers). 2. Central nervous system (CNS) involvement by CLL or a history of progressive multifocal leukoencephalopathy (PML). 3. Significant cardiovascular disease (e.g., symptomatic arrhythmias, congestive heart failure, myocardial infarction within 6 months prior to screening) or Class 3-4 cardiac disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Acalabrutinib (ATC Code: L01EL02) 100 mg twice daily (BID). **Route of Administration:** Oral (capsule, to be taken with 8 oz of water). **Duration of Treatment:** 48 cycles (28 days per cycle) or until disease progression, unacceptable toxicity, or death. Patients deriving clinical benefit may continue treatment beyond 48 cycles.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing CLL-related symptoms and co-morbidities. **Prohibited:** Live attenuated vaccines within 4 weeks of the first dose of study treatment; concurrent investigational or systemic anticancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Confirmation of IWCLL criteria, Baseline Labs
Baseline	Day 0	First Dose of Acalabrutinib, Quality of Life Questionnaires
Treatment Cycles	Every 28 Days (Cycles 1-48)	Safety Labs, Efficacy Assessment, Adverse Event Monitoring, PROs
End of Study	Follow-Up	Safety follow-up visit approximately 30 days from the last dose of study treatment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability profile characterized by the number of patients experiencing adverse events (AEs) and serious adverse events (SAEs). **Measurement Method:** Clinical assessment, physical examinations, and routine hematologic/chemistry laboratory evaluations.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) per IWCLL 2018 criteria. * Duration of Response (DOR) and Progression-Free Survival (PFS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient interviews, physical examinations, and hematological assessments recorded at every cycle. Common expected AEs monitored include headache, diarrhea, and cytopenias. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, particularly concerning cardiovascular events or severe infections (e.g., COVID-19 tracking was implemented during the pandemic). **Data Monitoring Committee (DMC):** Internal and independent review committees tracking safety signals globally.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving ≥ 1 dose of acalabrutinib); Efficacy Set (for ORR, PFS, and OS calculations). **Sample Size Justification:** Target $N=552$ is structured across 3 cohorts (≥ 300 treatment-naïve, ~ 200 relapsed/refractory, up to 40 prior ibrutinib therapy) to provide adequate power for safety and descriptive efficacy assessments within each subgroup. **Handling of Missing Data:** Standard survival analysis methodologies including right-censoring for PFS and OS.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of investigator sites to ensure protocol adherence and accurate data entry. **Audit Trail:** Secure eCRF systems utilizing data clarification forms to resolve discrepancies, with independent review facility verification for response criteria.

11. Study Identifiers & Governance

Primary ID: D8220C00008 **Registry Number:** NCT04008706 **Posting ID:** 517912925405500954 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ASSURE **Official Regulatory Title:** A Phase 3b, Multicenter, Open-Label, Single-Arm Study of Acalabrutinib (ACP-196) in Subjects with Chronic Lymphocytic Leukemia

12. Clinical Framework (CLL Specific)

Primary Condition: Chronic Lymphocytic Leukemia (CLL) **Disease Codes:** MeSH: D015451 | SNOMED: 51092000 | HPO: HP:0005550
Diagnosis Threshold: Active disease meeting IWCLL 2018 criteria for requiring treatment **Medical Methodology:** Targeted covalent Bruton Tyrosine Kinase (BTK) inhibition **Optimization Target:** Disease remission (ORR), progression-free survival, and manageable toxicity profile

13. Study Procedures & Methodology

Management Pathway: Daily continuous targeted oral antineoplastic therapy **Education Protocol (HCP):** Investigator training on IWCLL 2018 response criteria and AE management **Education Protocol (Patient):** Oral medication adherence instructions and AE reporting guidelines **Quality Audit Mechanism:** Independent response evaluation and safety signal tracking

14. Observation & Follow-Up Schedule

Total Duration: 48 treatment cycles (1 cycle = 28 days) **Onsite Visit Cadence:** Every cycle (Q4W equivalent) for the first 48 cycles **Remote Monitoring Frequency:** Supplemental telehealth visits (especially adapted during the COVID-19 pandemic timeframe) **Checklist Review Interval:** Symptom and PRO questionnaires (e.g., EORTC-QLQ-C30) reviewed continuously

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				